

EDUCATION

WORK EXPERIENCE

SKILLS

content: SKILLSProgramming Languages: Java, Python, C, Golang, Shell Script.Backend/Infra: Kubernetes, Docker, M

PROJECTS

PROJECTS

Mini Google - Distributed Search Engine [Java] <https://github.com/xuanhe95/noova>

Sep. 2024 - Dec. 2024

Designed and implemented a distributed search engine, including a TLS-encrypted HTTP Server for cross-node messaging, and a Coordinator-Worker Key-Value Store with sharding, versioning, replication, and consistency maintenance. Deployed across multiple AWS EC2 nodes for scalable and fault-tolerant processing.

Developed a Spark-inspired parallel computing engine with RDD abstractions and operator support. Built a distributed web crawler on top of it, capable of crawling and indexing 300K+ web pages. Ensured compliance with the robots exclusion protocol through adaptive request scheduling. Replaced disk I/O with in-memory tables for faster processing, achieving a throughput of 400+ pages/min.

Achieved <1s Query Latency with high accuracy, utilized inverted indexes with word segmentation and priority ranked index lists for efficient retrieval, followed by fine-grained ranking weighted by PageRank, TF-IDF, and term proximity. Implemented asynchronous retrieval and request pruning to improve efficiency.

Melody Hub - Music Database, [TypeScript] <https://github.com/xuanhe95/MelodyHub>

Feb. 2024 - May. 2024

Built a full-stack music database using React & Node.js, supporting music search by album, song title, or artist. Implemented user authentication, playlist management, and recommendations. Deployed on AWS with Docker.

Optimized performance of multi-table queries (100K+ rows each) through SQL optimization, indexing, and pruning, cutting latency from > 1min to < 2s. Used Redis and RabbitMQ to enhance system scalability and responsiveness. Designed and documented RESTful APIs with Swagger, enabling clean routing and simplified collaboration.